



Process Book

COM-480 Data Visualization

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Introduction

ChinaBev is an interactive data visualization project about China's freshly made beverage market. The project combines two datasets: a brand-level dataset covering 1,219 beverage brands, and a consumer behavior dataset containing 31,800 order records. Together, these datasets allow us to look at the market from two complementary angles: how beverage brands are positioned in terms of price, category, and store network, and how consumers behave across cities, age groups, motives, and consumption scenarios. Our goal was not only to present descriptive statistics, but to build a clear and engaging exploration tool for users who may not be familiar with Chinese beverage culture. The final website guides the user through three main perspectives: a personalized preference tree, a brand market landscape, and a consumer behavior analysis. This process book summarizes how our project evolved from the first sketches and exploratory ideas into the final interactive visualization. It also explains the main design decisions, technical challenges, changes from the initial milestone, and the contribution of each team member.

Project Path

Here is the path we took to bring our web project to life, highlighting our chronological progression, design iterations, and technical choices.

- Our journey began with a brainstorming session focused on our team's background. With two Chinese members in our group, we saw an opportunity to explore a cultural niche, something familiar to us but less known to an international audience. We initially mapped out several avenues, including interactive guides on regional Chinese tourist attractions, maps of local dialects, and visualizations of traditional folk customs. However, we quickly ran into data procurement difficulties. There were virtually no open-source, structured datasets for these highly localized topics in English repositories, and our initial web-scraping attempts yielded fragmentary data that lacked the depth needed for a solid data visualization project.
- The turning point came when we explored Chinese data science communities. There, we found a comprehensive dataset tracking the contemporary Chinese beverage market, covering consumer demographics, buying behavior across different regions, and market shares for hundreds of emerging drink brands. Despite its depth, the raw data required substantial preprocessing. To make it usable, we built a data-cleansing pipeline: we translated the linguistic schema and categorical tags into English, audited the tables to remove corrupt records, and filtered out rows with missing values.
- With a clean dataset, we clarified our target audience, project goals, and the specific data story we wanted to tell. When looking closer at our clean data, we realized it contained a great mix of macro market trends and micro consumer habits. This gave us the perfect opportunity to tailor our website to international consumers who want to explore modern Chinese tea and beverage culture. From our own experiences, we knew that outside of China, people

often feel completely lost when trying to order a drink because they cannot read the Chinese brand names, and do not understand the differences between countless milk tea chains. Our goal was to use data to remove this confusion.

- To guide the audience smoothly from being total beginners to making their first confident drink order, we designed a three-module framework to tell our data story step-by-step. First, the *Brand Market Landscape* acts as a macro introduction, showing users which brands are the most popular, their market shares, and how they position themselves in terms of pricing and store numbers. Second, the *Consumer Behavior Analysis* dives into the social trends, explaining when people usually buy these drinks, how much they spend on average, and how consumption habits change across different cities. Finally, the *Consumer Preference Tree* provides a fun, personalized conclusion, where users can interactively answer a few simple questions about their taste preferences to get a direct product recommendation.
- During the Milestone 2 phase, we translated this framework into a multi-page structure. Our initial blueprint featured a central landing homepage that introduced the cultural background and acted as a gateway to three separate sub-pages. At this stage, our team was highly ambitious, sketching out feature-heavy interfaces. For the market page, we planned an Interactive Bubble or Logo Cloud of the top 20 brands, where bubble radius represented store counts. For the behavior page, we designed a city pyramid structured by city tiers to show regional consumption habits. For the preference tree, we even envisioned a backend machine learning script to analyze a user's personal situation and predict their ideal drink matches. We believed packing the website with these diverse features would make it a comprehensive and exciting platform.
- Milestone 3 provided a vital reality check through peer and TA feedback. The critiques forced us to re-evaluate our layout and pivot toward a cleaner, more focused user experience. We realized that splitting the application into separate sub-pages broke the narrative flow and caused unnecessary clicking. To fix this, one of our teammates overhauled the core structure, rebuilding the web application's skeleton from the ground up. This modular refactoring allowed the rest of the team to code their respective visualizations simultaneously without git merge conflicts.
- Following this structural shift, we simplified our design. We consolidated the user journey into a single, seamless scrolling page, adding a sticky top-right navigation dock for quick jumping between sections. We stripped away redundant charts to reduce visual noise, redirecting our coding efforts toward perfecting our primary visualization components. Crucially, we re-ordered the information hierarchy to create a natural narrative. We moved the highly engaging and interactive *Consumer Preference Tree* to the very top as an immediate hook. Once users discovered their personalized drink type, they could scroll down into the more detailed analysis modules: the macro-level *Brand Market Landscape*

and the localized *Consumer Behavior Analysis*.

- With the individual visualization components working as isolated prototypes, our final stage focused on cross-component integration, state synchronization, and rigorous debugging. We established a unified root configuration for global CSS variables, ensuring standardized typography, absolute spacing scales, and synchronized theme palettes across all three independent modules. More importantly, during collaborative code reviews, we tackled significant timing bottlenecks within our asynchronous data pipelines. We optimized the asynchronous fetch queries and streamlined the event dispatching mechanisms. This ensured that whenever a user interacted with the global city filters, the data update event correctly cascaded across all dependent chart renderers simultaneously without layout shifting, race conditions, or duplicate DOM drawing. The final result is a highly responsive, technically stable web project that successfully translates complex datasets into a fluid, story-driven digital experience.

Initial Sketches and Plans

Our architectural goal is to map a macro-level “Shops Dataset” (1,219 brands) with a granular micro-level “Consumer Dataset” (31,800 transactions) across three interactive domains: (1) Brand Market Landscape, (2) Consumer Behavior Analysis, and (3) an interactive Recommendation Engine.

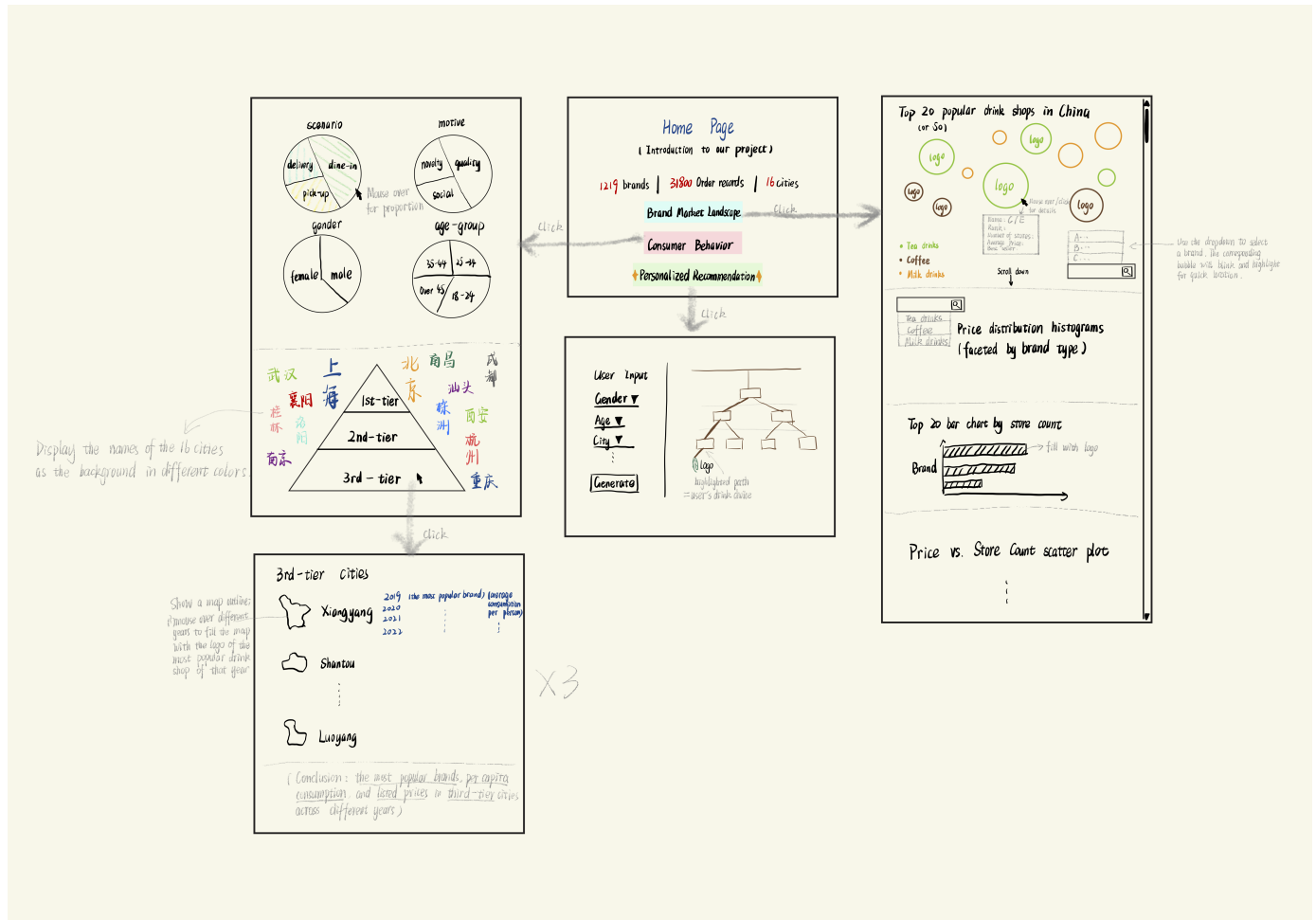


Figure 1: Initial interface layout sketch demonstrating coordinated multi-panel design.

As mapped out in early sketches (Figure 1), the multi-panel layout secures strict

visual coherence: the left split hosts demographic donuts and pyramids, while the right quadrant isolates the brand market overview.

Design Decisions

To maintain consistency and perceptual clarity across all three strategic domains of our dashboard, we implemented the following tailored visualization designs:

- **Domain 1 (Personalized Recommendation Tree):** Arranged an active hierarchical tree layout that traces interactive input configurations (age, gender, budget) down to specific brand leaves via path-highlighting nodes.
- **Domain 2 (Brand Market Landscape):** Standardized a distinct categorical color scheme (**Tea** in Amber Gold, **Coffee** in Emerald Green, **Milk Drinks** in Indigo Blue) to create an immediate mental shortcut for users navigating through competitive market tracks.
- **Domain 3 (Consumer Behavior Analysis):** Used synchronized pie and donut charts paired with custom glowing shadow accents. This helps viewers instantly compare regional demographic distribution shifts when toggling city filters.

Challenges Faced

SECTION 1: Decision Tree Product Type Prediction

The first challenge was converting the trained decision tree model into a structure that could be used directly for visualization. A standard model output is not convenient for a web interface, so we transformed it into a clear JSON format with explicit node IDs, split conditions, branches, leaf predictions, sample counts, and class distributions. This made the model reusable both for computing the recommendation and for drawing the full tree with D3.js.

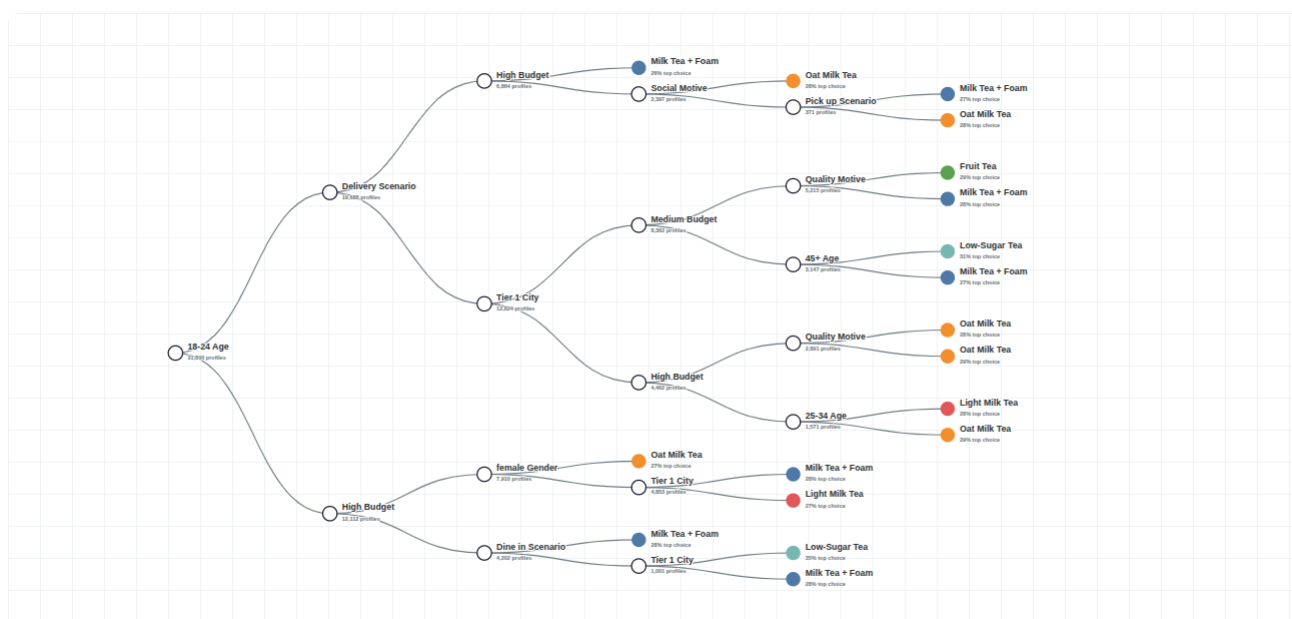


Figure 2: Initial full decision tree structure rendered from the exported JSON model.

The second challenge was highlighting the exact prediction path followed by the user's answers. To make this work, the JSON structure had to stay consistent across

both the traversal logic and the visualization logic. When the user selects a profile, the system follows the corresponding branches, stores the visited node IDs, and D3 redraws the tree with the active path emphasized. This avoids presenting the recommendation as a black box, since users can visually follow how their profile leads to the final product type.

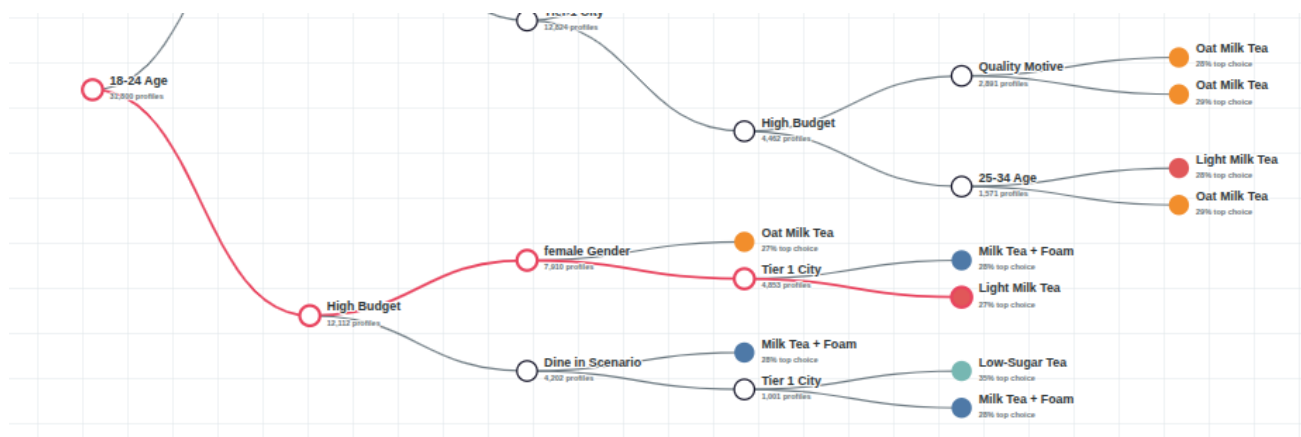


Figure 3: Highlighted prediction path showing how a user profile moves through the tree toward a final product type.

SECTION 2: Brand Market Landscape (Macro Overview)

The primary challenge stemmed from the sheer density and skewed scale of our dataset. Plotting 1,219 brands simultaneously created an illegible, overlapping mess of dots. The extreme magnitude difference, where massive national chains have 25,000+ stores while local boutique shops have only one, rendered linear charts useless, as 90% of the data collapsed into a solid black block at the bottom of the axis.

Furthermore, introducing a toggle between linear and logarithmic axes to fix this issue created a cognitive hurdle; users found it disorienting to see dots shift radically across the screen. I had to fine-tune the transition durations and axis tick generation in D3.js so that the structural scaling change felt mathematically smooth and intuitive rather than visually jarring.

SECTION 3: Consumer Behavior Analysis

Our multi-component architecture made real-time data sync quite tricky. Because clicking one city filter forced a massive update across four separate demographic charts and the calculation engine at once, we ran into race conditions. These asynchronous timing bugs occasionally caused the dynamic textbox to display data from the previously selected city instead of the new one. To fix this, I rewrote the event dispatching logic, creating a strict sequential pipeline that guaranteed data arrived in the correct order.

Another bottleneck appeared when trying to nail down the interface's tactile feel. Combining JavaScript-driven mouse tracking for the 1-to-5 slider with glow-heavy CSS filters caused noticeable frame drops when users moved their mice too quickly. I resolved the stuttering by debouncing the script-based event listeners and shifting the rendering workload onto hardware-accelerated CSS transitions, ensuring a lag-free user experience.

Final Visualization

The final web application successfully combines all our individual components into a single, cohesive interface. We built a clean, data-driven platform designed to help international users easily understand the complex Chinese beverage market and find the right drinks for themselves.

Our live project is deployed and publicly accessible. We warmly invite you to explore the interactive interface firsthand at the following link:

<https://china-bev.netlify.app/>

Changes from the First Milestone

Our final production interface evolved significantly from the static chart concepts proposed in Milestone 1. We introduced interactive visual models and synchronized component linkage across all three operational domains:

1. **Domain 1 (Personalized Recommendation Tree):** We replaced the standard demographic form-matching blueprint with an active, visual *Hierarchical Decision Tree Diagram*. Users can click custom radios to instantly trace how their age, budget, and scenario metrics traverse the network nodes to highlight a specific, tailored brand leaf in real time.
2. **Domain 2 (Brand Market Landscape):** Instead of plotting simple bar charts for store counts, we developed a force-directed *Bubble Cloud* featuring SVG circular clipped brand logos. Furthermore, we upgraded the static price scatter graph into a dual-mode switching viewport that lets users instantly swap between a Jittered Scatter view and a Grouped Frequency Histogram via live control buttons.
3. **Domain 3 (Consumer Behavior Analysis):** The initial plan relied on isolated, static demographic charts. We transformed this entire section into a reactive dashboard governed by global city-tier toggle pills. Clicking any urban tier dynamically recalculates and animates the donut slices while bringing up a custom text panel that presents comparative statistical variance.

Peer Assessment

Throughout this project, our team worked very closely together. We made sure all the important decisions, such as choosing the features and designing the layout of the website, were discussed and decided as a group. Everyone actively shared their ideas, which helped us solve problems and keep the story of our data connected. Once we agreed on the final structure, we divided the development work according to each person's skills. Here is the breakdown of our individual contributions:

SECTION 1 Decision Tree Preferred Product Type Prediction : Missipsa Annane
As the developer for the *Decision Tree Preferred Product Type Prediction* section, I was responsible for building the interactive recommendation module at the top of the website. I designed the user input form around six profile variables: city tier,

age group, gender, consumption motive, scenario, and budget. Based on these selections, the system traverses a precomputed decision tree and identifies the most relevant product type for the user profile. I also implemented the D3.js decision tree visualization. The full tree is rendered on page load with pan and zoom support, while the generated recommendation highlights the exact path followed by the user's answers. To make the model easier to understand, each node displays a readable split condition or predicted drink category, and hovering over a node shows extra information such as the profile size, split rule, suggested drink, most ordered drink, and top-choice share. Finally, I developed the result panel shown under the tree. After the user generates a profile, it summarizes the recommended product type, the most ordered drink among similar consumers, the product distribution, and the most popular brands for that profile. This section was designed to make the project immediately engaging for users, while also connecting the predictive model to concrete consumer behavior patterns from the dataset.

SECTION 2 Brand Market Landscape : Yujia Wang

As the developer for the macro-level *Brand Market Landscape* panel, she held full ownership over the technical delivery of the price scatter plots, segmented histograms, force-directed logo bubble clouds, and quadrant positioning matrices. She designed all filtering states, interactive adjusters, and axis toggle logic while coordinating global category rules with teammates to maintain application-wide CSS styling continuity.

SECTION 3 Consumer Behavior Analysis : Lingyi Zhu

As the developer for the *Consumer Behavior Analysis* panel, I was fully responsible for building the interactive demographics dashboard and its underlying data logic. I developed the four core consumer profile charts and implemented custom CSS hover effects featuring reactive neon glowing highlights. To elevate the user experience, I engineered a responsive calculation box that automatically compares city-tier data against the national average in real time. Additionally, I integrated the cross-category product popularity slider and designed the bottom summary container to successfully translate complex datasets into clear, readable market insights.